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March 31, 1983

MILITARY SPECIFICATION

BEEF SWISS STEAK, WITH GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers beef swiss steaks with gravy thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specifications For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Institutional Meat Purchase Specification for Fresh Beef Series 100

(Application for copies should be addressed to the Director, Livestock, Meat, Grain and Seed Division, 14th Street and Independence Avenue, SW, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

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2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of the following recognizable primal or sub-primal cuts of the round, shank-off (round, top round, bottom round and knuckles). Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have not more than a minor amount of lean, fat or bone from an adjacent cut. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade, and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, and/or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding the fresh chilled state for not more than 4 days after certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding raw beef in the frozen state for not more than 120 days in the freezer is permitted provided that the beef is:

- placed in the freezer within 4 hours after certification
- frozen to 0°F or lower within 72 hours after placement in the freezer
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

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3.2.2 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, and under shear stress. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.3 Tomatoes, crushed, canned,. Canned tomatoes shall be peeled, cored, mature, crushed tomatoes packed in tomato puree. The use of safe and suitable firming and acidification ingredients and salt is permitted. The canned tomatoes shall have not less than 8.0 percent tomato soluble solids and shall possess a red flesh color, normal character, and a good distinct acid sweet tomato flavor and odor. The crushed tomatoes shall be free of extraneous vegetable material and objectionable core material and skins (peel).

3.2.4 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.5 Sugar, white, granulated. The sugar shall be white, refined, granulated cane or beet sugar or a blend of both and shall comply with MIL-STD-900.

3.2.6 Oil, vegetable. The vegetable oil shall have a stability of 15 hours (minimum), a free fatty acid content not to exceed 0.05 percent by weight, and a moisture/volatile matter not to exceed 0.06 percent by weight and shall pass the cold test of 5.5 hours (minimum).

3.2.7 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.8 Onions, dehydrated, chopped. The dehydrated chopped onions shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for dehydrated Onion and Garlic Products.

3.2.9 Celery seed, ground. Ground celery seed shall be prepared from the seed of Apium graveolens L. The ground celery seed shall be light to rich brown in color and possess a characteristic celery odor and flavor with a warm to slightly bitter taste. The celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 55 sieve.

3.2.10 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that 95 percent shall pass through a U.S. Standard No. 16 sieve.

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3.2.11 Bay leaves, ground. Ground bay leaves shall be derived from the dried leaves of Laurus nobilis L. and shall possess a pleasant, aromatic odor and pungent, mildly bitter flavor with a pale green to yellow-green color. The volatile oil content shall be not less than 1.0 mL per 100 grams of ground bay leaves. A minimum of 95, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.12 Color, caramel. Caramel color shall comply with the Food Chemicals Codex.

3.2.13 Fat, frying. Frying fat shall be prepared from animal or vegetable (excluding coconut oil) fats, or a blend of animal and vegetable fats, which have been hydrogenated and deodorized. The frying fat shall have a clean bland flavor. The frying fat may be liquid or plasticized, may contain antioxidants in combination and amounts allowed by FDA, and shall contain antifoaming agents. The fat shall have a maximum free fatty acid content of 0.05 percent (oleic), a stability of not less than 40 hours (A.O.M.), and a smoke point of not less than 400°F.

3.2.14 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anticaking agent.

3.2.15 Water. Water used for formulation, ice making and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.16 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Preparation of beef swiss steaks. The steaks shall be formed to meet the requirements of IMPS Item No. 1102, USDA certified, and shall be approximately 6 ounces in weight and 0.5 inches in thickness.

a. Boned and trimmed meat meeting the requirements of 3.2.1.1 shall be further trimmed as necessary and in such a manner so as to assure compliance with finished product requirements.

b. The steaks shall be sufficiently cooked to brown the outer surfaces by any method except moist heat. Frying fat, if used, shall be filtered at least daily and the system cleaned at least once a week. The fat shall be discarded when foaming becomes a problem or when undesirable odors or flavors develop.

c. The browned steaks shall be rapidly cooled to 28°F to 40°F and held in this temperature range for not more than 24 hours prior to filling.

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d. Cooling of the browned steaks is not necessary if filling of tray pack cans occurs immediately after browning.

3.4 Preparation of the gravy. The gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	74.37
Tomatoes, crushed, canned	13.00
Starch	5.25
Onions, chopped, dehydrated	4.50
Oil, vegetable	1.00
Salt 1/	.85
Sugar	.50
Lecithin	.25
Caramel color	.10
Pepper, black	.07
Garlic powder	.05
Bay leaves	.03
Celery seed	.03

1/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

a. A slurry shall be made by mixing the starch, vegetable oil, and part of the water.

b. Any emulsion shall be made by mixing the remainder of ingredients (except the slurry). The emulsion shall be heated to approximately 140°F.

c. The slurry shall be added to the emulsion to form the gravy.

d. The gravy shall be heated to 180°F to 190°F and held in this temperature range for approximately 5 minutes

e. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

f. The gravy shall be filled within 4 hours after cooking. The prepared gravy shall be held in the temperature range of 150°F to 180°F at all times prior to filling into the tray pack cans.

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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with 12 intact beef swiss steaks shingled in two rows and covered with gravy such as to conform to the finished product requirements, and to the following requirements:

a. The temperature of the steaks at time of filling shall be in the temperature range of 28°F to 40°F, unless filling occurs immediately after browning.
b. The temperature of the gravy at time of filling shall be 150°F to 180°F.
c. Immediately after filling, each can shall be hermetically sealed under a vacuum of 5 to 6 inches of mercury as determined by presence of visible concavity of the can lid.
d. Each filled and sealed tray pack can shall be in the retort process within 1 hour after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 250°F or below until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.
b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.
c. There shall be no color foreign to the product. purple, blue, yellow or white.
d. Not less than nine of the beef swiss steaks in a can shall be distinct whole steaks.
e. Total weight of opaque connective tissue, ligament, tendon, fat chunks, blood vessel, cartilage, or glandular material shall be not more than 2 ounces.
f. No individual can shall contain a bone piece measuring more than 1/8 inch in any dimension.
g. No individual can shall have a steak with adherent fat or a fat pocket measuring 1/2-inch or greater in any dimension.
h. Steaks shall not be dry, tough, rubbery, or mushy.
i. Gravy consistency shall be smooth, without lumps and shall not be excessively thick or thin.
j. The average drained weight of the steaks shall be not less than 48.0 ounces.
k. The drained weight of the steaks in an individual can shall be not less than 44.0 ounces.
l. The average net weight shall be not less than 106 ounces.
m. No individual can shall contain less than 104 ounces of the product.
n. The average fat content of the finished product shall not be greater than 7.5 percent.

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o. The fat content of any individual tray pack shall not be greater than 9.5 percent.
p. The salt content of any individual tray pack shall be not greater than 1.25 percent or less than 0.5 percent.
q. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The meat component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or other applicable purchase document.

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4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

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TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral or any exposed lymph gland measuring 0.5 inch or more in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension.
	201	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	205	Presence of heavy connective tissue on lower edge of short plate or flank.
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more.
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	208	Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	211	Presence of dehydrated surface measuring 1.0 square inch or more.
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	214	Presence of cod, udder, kidney, or pelvic fat.

1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g.; glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the lot.

4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

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4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray packs shall be heated for 35 minutes in 140° to 150°F water, opened, and inspected for the defects listed in table IV.

TABLE III. Doubling sampling plan for product inspection 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	---	0	2
	8	16	1	2
3201 to 35000	13	---	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptable number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Less than 12 steaks in a can.
102	Less than 9 distinct whole steaks in a can.
103	Presence of bone measuring more than 1/8 inch in any dimension.
104	Drained weight of steaks in a can is less than 44.0 ounces. 3/
105	The average drained weight of steaks is less than 48.0 ounces. 3/
106	Steaks not shingled in two rows lengthwise.
107	Steaks dry, tough, rubbery, or mushy.

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TABLE IV. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
108	Total weight of opaque connective tissue, ligaments, tendons, fat chunks, blood vessels, cartilage or glandular material is more than 2 ounces.
109	Gravy consistency is not smooth, is lumpy, or is excessively thick or thin.
110	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by the AMS agent).

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify rendered fat and to remove gravy without breaking the steaks. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the steaks. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. Drained weight shall be reported to the nearest 0.1 ounce.

4.5.3.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analyses
- Three for submission to contractor
- Three for retention by an AMS agent as a reserve sample.

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The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Methods of Analyses of the Association for Official Analytical Chemists, chapter: Meat and Meat Products, section: Meat, except that preparation of the sample shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:

a. The average fat content of three the tray packs is greater than 7.5 percent.
b. The fat content of any individual tray pack is greater than 9.5 percent.
c. The salt content of any individual tray pack is less then 0.5 percent or greater than 1.25 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with 21 CFR, Part 113, Subpart D, and the can manufacturer's requirement.

4.5.6 Vacuum examination. Cans shall be examined 24 hours after sealing to determine vacuum retention. To examine, lay a straight edge from bead to bead and verify that the distance from the straight edges to the lid is greater at the middle of the can than near the bead.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Interior packing with fiberboard liner or pads not as specified.
Arrangement or number of cans not as specified.

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4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4. and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a fiberboard box, constructed and closed in accordance with style HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The inside box dimensions shall be 12-3/4 inches in length by 10-1/2 inches in width by 8-1/2 inches in depth. The cans shall be packed flat, four in depth within the box with two cans placed with covers down and two cans with covers up. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3s fiberboard. The dimensions of the fiberboard pads shall be 11-1/4 inches by 9-1/4 inches (± 1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

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5.2.3 Level C packing Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.5.4 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge sealed can in boiling water. Bring to boil, continue boiling for 40-45 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. CAUTION: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F. WARNING: Do not place closed can in oven as heating will cause the can to burst.

YIELD

Approximately 12 servings of one steak each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

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5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-A514

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44227	Beef Swiss Steak, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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